Chapter 6 — Bioenergetics

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Oxygen released in photosynthesis comes from: (a) Carbon dioxide (b) Glucose (c) Water (d) ATP
2. Non-cyclic photophosphorylation produces: (a) ATP only (b) ATP, NADPH and O2 (c) NADPH only (d) Glucose only
3. Calvin cycle occurs in the: (a) Stroma (b) Thylakoid lumen (c) Matrix (d) Cytoplasm
4. CO2-fixing enzyme in Calvin cycle is: (a) Pepsin (b) ATP synthase (c) Rubisco (d) Hexokinase
5. Glycolysis occurs in the: (a) Mitochondrial matrix (b) Nucleus (c) Thylakoid (d) Cytoplasm
6. Krebs cycle occurs in the: (a) Thylakoid (b) Cytoplasm (c) Mitochondrial matrix (d) Inner membrane
7. Final electron acceptor in aerobic respiration is: (a) NAD+ (b) CO2 (c) Oxygen (d) FAD
8. Aerobic respiration net ATP (textbook) is: (a) 36 ATP (b) 4 ATP (c) 18 ATP (d) 2 ATP

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. How are light-dependent and light-independent reactions linked?
2. Differentiate between cyclic and non-cyclic photophosphorylation.
3. Why is glycolysis common to both aerobic and anaerobic respiration?
4. How does the Krebs cycle contribute to ATP production?
5. Why is oxygen essential for aerobic respiration?
6. How does fermentation allow glycolysis to continue in the absence of oxygen?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain the light-dependent reactions of photosynthesis. **(4)**
(b) Describe the Calvin cycle. **(4)**
Q.4 (a) Describe the stages of aerobic respiration. **(4)**
(b) Explain the electron transport chain and chemiosmosis. **(4)**

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